Do Trump's Tweets Spur Market Volatility?

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Trump's tweets and the market

- Gjerstad et al. (2021) find that the tweets are followed by higher uncertainty, higher trading volume and a decline in the US stock market, strongest for tariffs and trade.
- Perico Ortiz (2023) finds that tweets on foreign policy and trade, monetary policy and immigration raise the CBOE Volatility Index (VIX), peaking about 2 hours after posting.
- With minute-level data on the stocks Trump named, Machus et al. (2022) find abnormal returns, higher trading volume and higher investor attention already before the tweets. This is consistent with the tweets commenting on events that had drawn attention.
- Abdi et al. (2023) classify about 1,200 of his economy-related tweets and find that 80% react to prior market developments rather than convey new information. The exception is tweets on the trade war, where he had direct policy authority.

Research question

Question

1. Is the daily log change in the VIX associated with the sentiment or topic of Donald Trump's tweets from January 2016 to January 2021?
2. Do the tweets carry new information on a daily setting?

Finding

1. Sessions with a tweet of extreme sentiment show a log change in the VIX 0.95% higher than in other sessions, but the difference does not extend to later sessions.
2. Among the 19 topics extracted by BERTopic, only a few coefficients are significant, and their signs change across horizons.
3. Coronavirus tweets came after the VIX had risen and election fraud tweets as it fell, consistent with tweets that comment on news rather than create it.

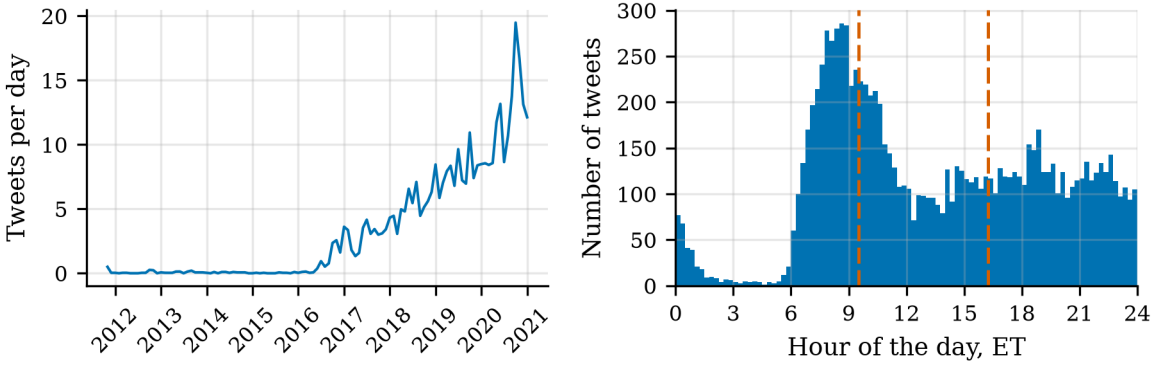
Data

- The tweets from @realDonaldTrump are taken from a dataset published on Kaggle, with times converted to Eastern Time (ET).
- The sample holds 10,566 tweets from 3 January 2016 to 8 January 2021 (Appendix A.1).
- We proxy market volatility by the VIX, the 30-day implied volatility of the S&P 500.
- V_t is the close of session t and $r_t = 100 \log(V_t / V_{t-1})$ its daily log change.

	Mean	Median	SD	Min	Max	Skewness	Excess kurtosis
V_t	17.7	14.7	8.82	9.14	82.7	2.83	11.9
r_t	0.0092	−0.72	8.25	−30.0	76.8	1.51	9.07

Summary statistics of the VIX close V_t and its daily log change r_t.

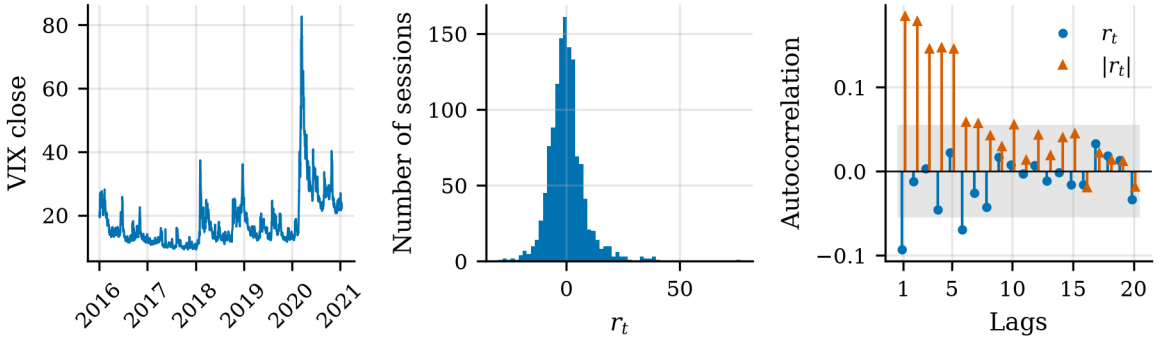
Tweets



Monthly mean tweets per calendar day (left) and time of posting for tweets from 2016 onward (right), with the open at 9:30 and the VIX close at 16:15 ET marked.

- Few tweets in the file precede 2016, so the sample starts on 3 January 2016.
- Posting peaks in the morning before the open and continues late into the evening.

Volatility



VIX close V_t (left), histogram of the daily log change r_t (middle), and autocorrelations of r_t and $|r_t|$ up to 20 lags (right).

- V_t is persistent, spiking in February and December 2018 and peaking on 16 March 2020.
- r_t is right-skewed and heavy-tailed, with a negative first-order autocorrelation.
- The autocorrelation of $|r_t|$ is significantly positive over the first 7 lags.

Trading sessions

$$\tau(i) = \min\{t : \bar{T}_t > T_i\}, \qquad \mathcal{T}_t = \{i : \tau(i) = t\}$$

- T_i is the time of tweet i and $\bar{T}_t$ the VIX close at 16:15 ET on the date of session t.
- Because the VIX closes at 16:15 ET and many tweets are posted outside trading hours, a tweet cannot simply be matched to its calendar date.
- Tweet i belongs to session $\tau(i)$, the first session whose VIX close follows it.
- In a week without holidays, a tweet on a Friday night and a tweet on the following Sunday both belong to the Monday session.

Of the 10,566 tweets, 23.1% are posted on trading days before the open, 23.1% between the open and the VIX close, 24.1% after it and 29.8% on weekends and holidays (Appendix A.1).

Sentiment scores

- We score the sentiment of each tweet with the Valence Aware Dictionary and sEntiment Reasoner (VADER) model of Hutto and Gilbert (2014).
- VADER is a lexicon of words rated by human judges from -4 to $+4$, adjusted by rules for capitalization, degree modifiers, negation, contrastive conjunctions and punctuation.
- It returns a compound score $s_i \in (-1, 1)$ for each tweet i.
- A threshold of 0.5 already classifies 58% of the tweets as positive or negative, so we mark a tweet as extreme only when $|s_i| \geq \theta$ with $\theta = 0.9$.

	Positive	Neutral	Negative
$\theta = 0.5$	3,658	4,389	2,519
$\theta = 0.9$	748	9,414	404

Numbers of the 10,566 tweets with $s_i \geq \theta$, $|s_i| < \theta$ and $s_i \leq -\theta$ at two thresholds.

Event regression

$$D_t = \mathbf{1}\Big\{ \max_{i \in \mathcal{T}_t} |s_i| \geq \theta \Big\}$$

$$r_{t+k} = c_k + \phi_k \, r_{t+k-1} + \gamma_k D_t + \varepsilon_{k,t}, \qquad \text{for each horizon } k \in \{-3, \ldots, 3\}$$

- $D_t = 1$ marks an event session, one that carries at least one extreme tweet.
- c_k is a constant, ϕ_k the coefficient of the lagged change and $\varepsilon_{k,t}$ an error.
- The lagged change absorbs the negative autocorrelation of r_t.
- We estimate the regression by ordinary least squares with the heteroskedasticity- and autocorrelation-consistent standard errors of Newey and West (1987) at 5 lags.
- Since tweets and volatility may both respond to the news of the day, γ_k measures an association rather than a causal effect.

Sentiment regression results

	$k=-3$	$k=-2$	$k=-1$	$k=0$	$k=1$	$k=2$	$k=3$
ϕ_k	-0.09^{***}	-0.09^{***}	-0.09^{***}	-0.09^{***}	-0.09^{***}	-0.09^{***}	-0.09^{***}
γ_k	-0.23	0.80^{*}	-0.42	0.95^{**}	0.25	0.40	-0.51

Sentiment regression results (the constant is not reported). Significance under the standard errors of Newey and West (1987) at $^{*}p \leq 0.1$, $^{**}p \leq 0.05$ and $^{***}p \leq 0.01$.

- In an event session, the log change in the VIX is 0.95% higher than in other sessions, which corresponds to 0.14 index points or 1.9% in variance.
- None of $\gamma_1, \gamma_2, \gamma_3$ is significant, so there is no evidence that a tweet of extreme sentiment predicts the VIX beyond the session that first reflects it.

Topic models

- The Latent Dirichlet Allocation (LDA) model (Appendix B.1) may generate topics that are challenging to interpret. This is mainly because it relies on a bag-of-words model that ignores word order and assumes that topics are independent.
- BERTopic retains the original text structure and better captures the semantic nuances and context of the text.
- We therefore extract the topics with BERTopic, following Grootendorst (2022).

Topic extraction

$$\text{tweet } i \xrightarrow{\text{all-MiniLM-L6-v2}} e_i \in \mathbb{R}^{384} \xrightarrow{\text{UMAP}} u_i \in \mathbb{R}^3 \xrightarrow{\text{HDBSCAN}} z_i \in \{-1, 0, \ldots, 18\}$$

- The pretrained sentence transformer embeds each tweet as a unit vector e_i.
- Uniform manifold approximation and projection (UMAP) maps e_i to u_i.
- Hierarchical density-based clustering (HDBSCAN) groups the u_i into clusters of at least 80 tweets, with core distances over neighborhoods of 40 other points.
- z_i is the topic of tweet i. Tweets not associated with any topic form an outlier class $z_i = -1$, which is disregarded in the subsequent analysis.

BERTopic topics

j	Label	Tweets	Representative words
0	Russia investigation	987	russia, mueller, collusion, fbi, investigation
1	News media	862	fake news, media, cnn, foxnews, nytimes
2	Border and immigration	655	border, wall, immigration, mexico, daca
3	Impeachment	580	impeachment, pelosi, schiff, ukraine, whistleblower
4	Tributes and greetings	510	god, nation, honor, veterans, happy
5	Rallies and endorsements	396	rally, crowd, endorsement, florida, pennsylvania
6	Thanks and short remarks	367	thank, thanks, maga, nice, true
7	Election fraud	351	election, fraud, rigged, ballots, mail-in
8	Economy	334	economy, stock, market, jobs, unemployment
9	Coronavirus	293	coronavirus, testing, cases, vaccine, ventilators
10	Protests and policing	252	anarchists, police, mayor, portland, cuomo
11	China trade	231	china, trade, tariffs, farmers, deal
12	Democratic candidates	228	biden, bloomberg, bernie, debate, sleepy
13	Middle East	217	israel, isis, syria, turkey, kurds
14	North Korea	150	korea, kim, summit, denuclearization, missile
15	Health care and tax reform	141	obamacare, repeal, preexisting, tax, cuts
16	Iran	118	iran, nuclear, deal, sanctions, iraq
17	Supreme Court	87	supreme, court, justices, judges, pack
18	Georgia election	84	georgia, kemp, signature, verification, ballots

BERTopic topics with their numbers of tweets and five representative words.

Topic regression

$$N_{j,t} = \big|\{i \in \mathcal{T}_t : z_i = j\}\big|, \qquad j = 0, \ldots, 18$$

$$r_{t+k} = c_k + \phi_k \, r_{t+k-1} + \boldsymbol{\beta}_k^\top \mathbf{X}_t + \varepsilon_{k,t}, \qquad \text{for each horizon } k \in \{-3, \ldots, 3\}$$

- To measure how each subject relates to volatility, we replace the sentiment indicator D_t with topic counts.
- $N_{j,t}$ is the number of tweets of session t on topic j, with z_i the topic of tweet i.
- $\mathbf{X}_t = (N_{0,t}, \ldots, N_{18,t})^\top$ collects the counts and $\boldsymbol{\beta}_k$ their coefficients.

Topic regression results, topics 0 to 8

j	Label	$k=-3$	$k=-2$	$k=-1$	$k=0$	$k=1$	$k=2$	$k=3$
	ϕ_k	-0.09^{***}	-0.10^{***}	-0.09^{***}	-0.09^{**}	-0.10^{***}	-0.09^{***}	-0.10^{***}
0	Russia investigation	-0.26^{**}	0.01	-0.22	0.30^{*}	0.25^{**}	0.00	0.22
1	News media	0.09	-0.07	-0.17	-0.31	-0.28	-0.07	0.02
2	Border and immigration	-0.12	0.07	-0.15	0.39^{**}	-0.06	-0.08	-0.34^{**}
3	Impeachment	-0.19	-0.14	0.11	0.31^{**}	0.26^{**}	0.14	-0.07
4	Tributes and greetings	0.08	0.40	-0.46	0.62	-0.56	-0.18	-0.17
5	Rallies and endorsements	0.21	0.72^{*}	0.11	0.36	0.47	-0.33	-0.43
6	Thanks and short remarks	0.35	0.03	-0.16	0.20	-0.53^{*}	0.12	0.17
7	Election fraud	-0.55^{**}	-0.29	-0.43^{**}	-0.33^{*}	-0.25^{*}	0.02	-0.03
8	Economy	-0.18	0.31	-0.73^{**}	-0.03	0.19	0.17	0.27

Topic regression results for topics 0 to 8. Significance at $^{*}p \leq 0.1$, $^{**}p \leq 0.05$ and $^{***}p \leq 0.01$.

- In the tweet session $k = 0$, a tweet on the border and immigration or on impeachment coincides with a log change in the VIX higher by 0.39% and 0.31%, respectively.
- Before the tweet session, the coefficients significant at the 5% level are all negative.

Topic regression results, topics 9 to 18

j	Label	$k=-3$	$k=-2$	$k=-1$	$k=0$	$k=1$	$k=2$	$k=3$
9	Coronavirus	−0.18	0.53	0.15	0.15	0.61*	0.07	−0.62
10	Protests and policing	−0.17	0.19	−0.22	0.40	−0.13	−0.10	0.44
11	China trade	0.32	−0.34	0.57	0.03	−0.46	−0.44	−0.39
12	Democratic candidates	0.21	−0.08	0.28	−0.47	−0.05	0.56	0.99
13	Middle East	−0.04	−0.28	−0.15	0.14	0.13	−0.68**	−0.89**
14	North Korea	−0.05	0.17	−0.34	0.65	0.25	0.50	0.30
15	Health care and tax reform	−0.50	−0.61	0.67*	0.83	0.45	−1.17***	−0.37
16	Iran	−0.19	−0.44	0.30	−0.46	−0.39	0.07	−0.30
17	Supreme Court	0.32	−0.05	0.23	0.24	0.14	−0.73	−0.31
18	Georgia election	0.26	−0.07	0.66*	0.89**	0.55*	0.06	−0.23

Topic regression results for topics 9 to 18. Significance at $^{*}p \leq 0.1$, $^{**}p \leq 0.05$ and $^{***}p \leq 0.01$.

- In the tweet session, a tweet on the Georgia election coincides with a log change in the VIX higher by 0.89%.
- The coronavirus and China trade topics show no relation at the 5% level.

Timing of the tweets

- China trade The China trade topic is not significant at any horizon, while Gjerstad et al. (2021) and Perico Ortiz (2023) find responses to it within hours of posting.
- Since the VIX is forward-looking, this pattern is consistent with the information in the tweets, if any, being priced in by the close of the session.
- Election fraud Most of its tweets follow the election of 3 November 2020, after which the VIX fell from 35.6 to 21.6 by 8 January 2021.
- Their negative coefficients are consistent with a fall in implied volatility as uncertainty about the outcome resolved.
- Coronavirus Of its tweets, 77% were posted after the VIX peaked on 16 March 2020, so they mostly followed the rise in volatility rather than led it.

- Overall, the timing of the tweets points to tweets that react to news rather than create it. This is in line with the reactive channel of Machus et al. (2022) and Abdi et al. (2023).

Conclusion

- We examine the daily association between Donald Trump's tweets and the VIX from January 2016 to January 2021, using VADER sentiment scores and BERTopic topics.
- Sessions with an extreme tweet show a log change in the VIX 0.95% higher than in other sessions, about 0.14 index points, but the difference does not extend to later sessions.
- Only a few topic coefficients are significant, and the timing of the coronavirus and election fraud tweets is consistent with tweets that comment on news rather than create it.
- A daily design can hardly separate the response to a tweet from the response to the news it comments on, since tweets and news both accumulate while the market is closed.

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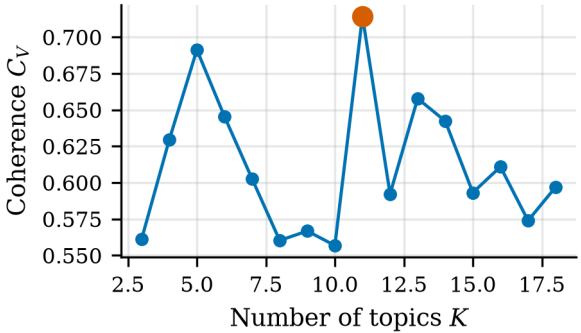
Code and replication data are available at *GitHub*.

Appendix A.1. Sample and trading sessions

Panel B. Sample, 3 January 2016 to 8 January 2021	
Tweets	10,566
Tweets per calendar day	5.76
Posted on trading days before the open at 9:30 (%)	23.1
Posted on trading days between the open and the VIX close at 16:15 (%)	23.1
Posted on trading days after the VIX close (%)	24.1
Posted on weekends and holidays (%)	29.8
Panel C. Trading sessions, 4 January 2016 to 14 January 2021	
Sessions	1,268
Sessions with a tweet	1,095
Tweets per session with a tweet, mean	9.65
Tweets per session with a tweet, median	7

The tweet file and the trading sessions (ET throughout).

Appendix B.1. Coherence and topics of LDA



Coherence C_V of LDA against the number of topics K.

Topic	Tweets	Most frequent words
0	9	great, thank, california
1	18	witch, hunt, people
2	21	great, honor, life
3	163	great, people, america
4	95	america, nation, great
5	10	america, billion, dollar
6	36	national, great, border
7	13	china, death, penalty
8	352	great, america, win
9	13	attack, racist, washington
10	5	vote, illegal, moore

Topics of the LDA model with eleven topics.

- We fit LDA to the 735 tweets of the word clouds with $K = 3, \ldots, 18$ topics.
- The coherence is highest at $K = 11$, with 0.714.
- Six of the eleven topics are labeled by *great*, and two topics hold 515 of the 735 tweets.

Appendix C.1. Words of the most extreme tweets of each session





Words from the most positive tweet of each session with $s_i \geq 0.9$ (left) and the most negative tweet of each session with $s_i \leq -0.9$ (right), with size increasing with frequency.

- Positive tweets center on {*great*, *thank*, *win*, *america* and *people*}.
- Negative tweets center on {*bad*, *people*, *fake news*, *democrat*, *witch hunt* and *crime*}.